InsightPilot: As-built Status of the System Architecture

Status as of November 12, 2026 · extends Figure 1 · CMPE 295A + 295B scope

DEPLOYMENT: ONE ORGANISATION · SELF-HOSTED / DOCKER COMPOSE

PRESENTATION TIER

Investigation Workspace
DONE
Ask questions; follow-ups inherit context
React · TypeScript · FastAPI

Plan Review & Approval
DONE
Clarify ambiguity; edit and approve plans
React · typed plan DAG

Lineage Inspector & Export
DESIGNED
Inspect / replay hops; analyst sign-off
React · self-contained artifact
Spec: UX screens 7 and 9

question + context / analyst-approved plan
clarification / plan / answer + evidence

ORCHESTRATION TIER: CUSTOM PYTHON 3.12

Semantic Resolver
DESIGNED
Bind terms to columns / grain
Halt and ask when ambiguous
BM25 · embeddings · rank fusion
Spec: deep dive 1

Plan Synthesiser
DONE
Generate typed DAG; validate sources, inputs and cycles
Python · constrained generation

Orchestrator & Replanner
DONE
Route all handoffs; own state
Budget / retry caps; replan [295B]
Custom Python · DAG execution

Model Client
DESIGNED
Plan / SQL / prose model calls
Schema-only or filtered samples
Provider-agnostic HTTPS API
Spec: D05

HTTPS

EXTERNAL MODEL PROVIDER
Frontier: planning / SQL
Small model: prose / chart choice
Organisation's own API key
Question + schema; optional samples
PII-filtered / schema-only mode

Outside the deployment
External service · no status

typed tasks + input references + budget
results + declared grain + lineage

SPECIALIST AGENT TIER: ALL TASKS AND RESULTS ROUTED THROUGH ORCHESTRATOR

Fetch / Integration Agent
DESIGNED
Generate / repair queries
Retrieve and prepare data
Python · SQLAlchemy · psycopg
Spec: deep dive 3

Analytics Agent
DESIGNED
Joins, KPIs, comparisons
Trends and aggregation
SQL · pandas
Spec: D03 handoff contract

Visualization Agent [295B]
PLANNED
Choose charts; return chart specification and rationale
Python · chart rendering
CMPE 295B

ML Agent [295B]
PLANNED
Time-boxed baseline models
Model card + uncertainty
scikit-learn · LightGBM
CMPE 295B

per-hop outputs / verification evidence
pass / warning / failure to orchestrator

VERIFICATION & LINEAGE TIER: CHECKS USE SQL / COMPUTE, NEVER LLM SELF-CRITIQUE

Hop Verifier
DESIGNED
Counts, grain, coverage, nulls
Totals and type / range checks
Independent SQL · local compute
Spec: deep dive 4

Cross-source Reconciler
PARTIAL
Profile join keys and overlap
Propose mappings for approval
DuckDB SQL · pandas
Profiling done Nov 10; proposals S4

Lineage Recorder & Replayer
DESIGNED
Record query, grain, checks and skips
Replay one hop; report differences
Python · PostgreSQL · local files
Spec: deep dive 6 · D04

independent SQL / profiles over sources
schema / row counts / coverage / grain

INTEGRATION TIER: READ-ONLY SOURCES; SEMANTIC DEFINITIONS TAKE PRECEDENCE

PostgreSQL Connector
DONE
Describe, sample, execute, profile
Inherit source read permissions
SQLAlchemy · psycopg

CSV / Excel Connector
DONE
Query uploaded files
Infer schema; profile without ETL
DuckDB · Python

REST JSON Connector [295B]
PLANNED
Third source type
Read-only API access
Python · HTTP / JSON
CMPE 295B

Semantic Layer [295B]
PLANNED
Definitions override inference
dbt semantic layer
Separate semantic interface
CMPE 295B

durable bindings, state, lineage and audit
replay metadata / evaluation outcomes

DATA, AUDIT & EVALUATION TIER

Application Store
DESIGNED
Org bindings, plans, investigations
Hop lineage, overrides, sign-off
PostgreSQL 16
Spec: D04 schema

Artifact Store
DESIGNED
Local step outputs and exports
Recorded statements / parameters
Local filesystem
Spec: Chapter 5

Append-only Audit Log
DESIGNED
Actor, source, statement, timestamp
Readable without application
JSONL · local filesystem
Spec: D06 · UX screen 12

Offline Evaluation Harness
DESIGNED
Ground truth + seeded errors
Accuracy and verifier coverage
pytest · question-set fixtures
Spec: D08

Done (implemented) · Partial: part built, rest scheduled · Designed: specification complete · Planned (CMPE 295B)

SUMMARY: 6 of 18 CMPE 295A components done; 1 partial; 11 designed with specifications; 4 planned for CMPE 295B.

READING: Tiers, cards, arrows and technologies are unchanged from Figure 1; only the status fill, badge and specification line are added.

Specifications: runs/insightpilot/tech/deep_dives.md, tech/architecture/D01–D10 and product/ux_spec.md in the project repository.